CyberShield: Secure File Encryption and Password Strength Analysis System

Project Title

CyberShield: Secure File Encryption and Password Strength Analysis System

Project Name

CyberShield

Objective

Develop a cybersecurity application for password strength analysis, secure file encryption/decryption, and cybersecurity awareness.

Problem Statement

Weak passwords and unencrypted files expose users to cyber threats. This project improves data protection and awareness.

Research Findings

Weak passwords are a common cause of breaches. Encryption protects data from unauthorized access.

Methodology

Requirement Analysis, Design, Development, Testing, Deployment, Maintenance.

Tools Used

Python, Tkinter, SQLite, cryptography, hashlib, VS Code.

Implementation

Login, Password Checker, File Encryption/Decryption, Cyber Tips modules.

Screenshots

Insert screenshots of Login, Dashboard, Password Checker, Encryption, Decryption, Tips pages.

Results

Application successfully checks passwords and encrypts/decrypts files.

Challenges

Encryption concepts, GUI design, key management, testing.

Future Scope

MFA, cloud storage, AI password suggestions, malware detection.

Conclusion

CyberShield demonstrates practical cybersecurity techniques for securing user data.